

The one-step-ahead predictive over time: process variance, PIT and variance decomposition

Claude

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Summary

This note follows the one-step-ahead predictive of the filter year by year, with and without bias correction. It looks at three things: the learned process variance, the PIT, and a decomposition of the predictive variance into process error and the other sources of variance. It asks whether the PIT flattens as the filter learns a smaller process variance. It is a companion to `claude-forecast-width-10.8.26`, which explains why the forecasts are too wide. It reaches five conclusions.

1. **With bias correction, the process variance falls steadily, by 20–50 times by 2017.** The median falls from 0.61 in 1991 to 0.01 (GEDI) and 0.03 (LandTrendr). Without correction it levels off at 0.1–0.25 after about 1995.
2. **The PIT flattens only a little, and only for GEDI.** For GEDI heteroskedastic R the PIT interquartile range (IQR) widens from 0.22 in 1995 to 0.28 by 2000 and then stays at 0.29–0.30 (calibrated: 0.5). For LandTrendr it does not widen at all (about 0.15–0.21), even though the process variance falls nine-fold over the same years.
3. **The bias correction centres the PIT; it does not flatten it.** Corrected, the mean and median PIT sit at about 0.54 in every year. Uncorrected they sit at about 0.27–0.36.
4. **The process variance stops mattering after the first few years.** The process share of the forecast variance is already small by the mid-1990s, so the remaining width comes from the observation error R , as shown in the forecast-width note.
5. **Process error is 1–4% of the predictive variance (median over sites) after 1995 with correction** (Section 3). The rest is the observation error R , the state uncertainty propagated from the previous analysis, and the disturbance part (the low tail from the disturbed mixture components). For LandTrendr, R is 64–84% of the variance. For GEDI heteroskedastic R , the disturbance part is the largest share (43–72%). That tail inflates the variance but not the hump in the PIT.

1 Runs and definitions

- **Runs.** The two full grids, without the disturbance product (so no year is flagged), at all 538 sites, for each of the four observation-error models:

- uncorrected: `experiments/fullRunNoBias`;
- bias-corrected: `experiments/fullRunBias`, with `bias = list(rho = 0.9, denom = "N", skip = 1)`.

Year t runs from 1991 to 2017. Year 1990 is excluded because its forecast starts from the population-wide prior on the initial state.

- **Process variance** v_{t-1} . The posterior mean of the process variance after assimilating y_{t-1} :

$$v_{t-1} = \mathbb{E}[1/\tau \mid y_{1:t-1}] = \sum_g \pi_{t-1}(g)/\tau_g,$$

where $\pi_{t-1}(g)$ is the posterior weight of grid point τ_g (`hist$procVar` at $t-1$). This is the process variance used by the forecast of y_t , so it is plotted at year t , aligned with that year's PIT. (`claude-forecast-width-10.8.26` plots v_t , the same series one year later.)

- **PIT** u_t . The probability integral transform of y_t under its one-step-ahead predictive: the predictive probability of a value $\leq y_t$ (`hist$pit`). If the predictive is calibrated, u_t is uniform on $[0, 1]$: mean and median 0.5, quartiles 0.25 and 0.75, IQR 0.5. An IQR below 0.5 means the PIT is concentrated in the middle, i.e. the predictive is too wide. A median away from 0.5 means it is biased (below 0.5: forecasts too high).
- **Across-site summaries**. For each year, the median, IQR (25th–75th percentile) and mean over the 538 sites, of v_{t-1} and of u_t .
- **Coverage error of the central 50% interval**. The fraction of sites with $|2u_t - 1| \leq 0.5$ (the observation falls in the predictive's central 50% interval), minus 0.5. Positive values mean the forecast is too wide; 0 means calibrated.
- **Periods**. 1991–95, 1996–2000, 2001–05, 2006–11 and 2012–17, for the PIT histograms in Figure 2.

2 Process variance and PIT over time

Process variance (Figure 1, left).

- **With correction it keeps falling**. The median falls fastest in the first five years and keeps falling slowly after that. For GEDI it reaches about 0.01, for LandTrendr about 0.03 by 2017.
- **Without correction it levels off**. After about 1995 the uncorrected median stays at 0.1–0.25, and for GEDI it creeps up: the filter uses process variance to absorb the forecast bias (as in the forecast-width note).
- **The mean is much larger than the median and levels off at 0.06–0.13 after 2000**. The distribution over sites is right-skewed: most sites learn a small process variance, and a minority with large year-to-year jumps learn a much larger one. For GEDI heteroskedastic R the upper quartile rises again after about 2013 (0.035 to 0.071), from sites whose biomass changes quickly late in the record.

Process variance and PIT over time, no disturbance product, 538 sites

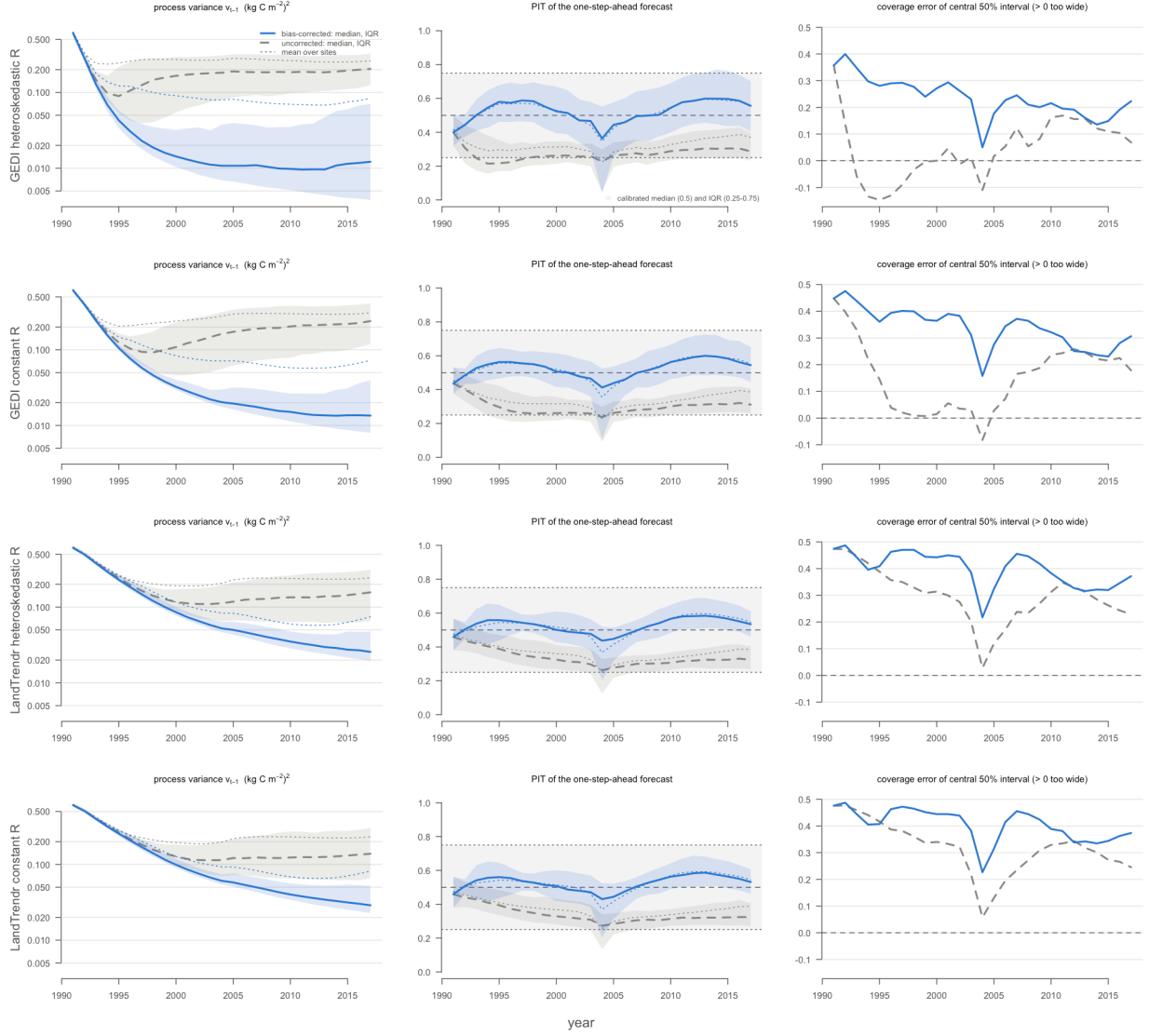


Figure 1: Per year, one row per observation-error model; blue = bias-corrected, gray dashed = uncorrected. *Left:* process variance v_{t-1} (log scale), median (thick line), IQR (band) and mean (dotted) over sites. *Middle:* PIT u_t , median, IQR and mean over sites. The shaded strip and dotted lines at 0.25 and 0.75 are the calibrated quantiles. *Right:* coverage error of the central 50% interval.

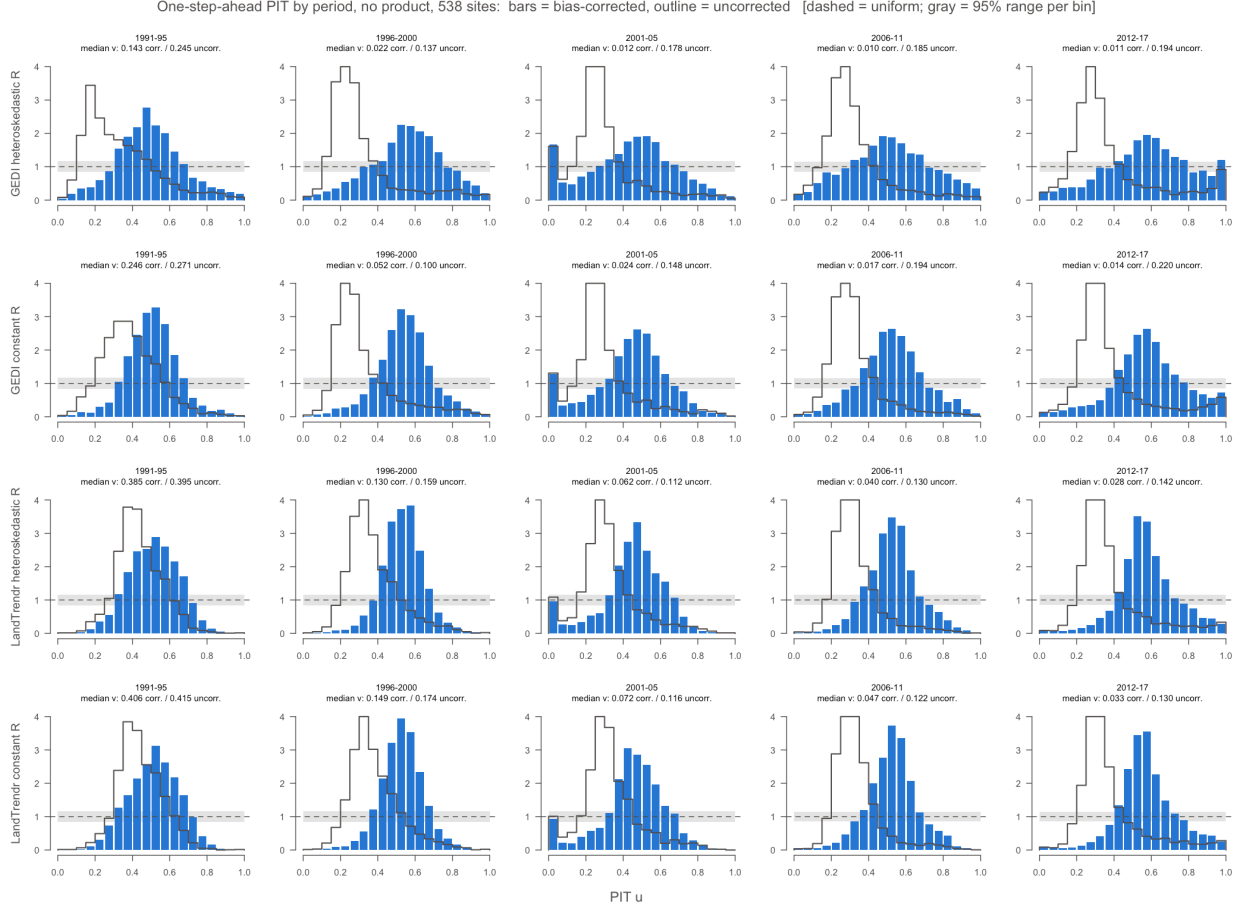


Figure 2: PIT histograms by period (all sites and years in the period). Bars: bias-corrected; outline: uncorrected; dashed: uniform; gray: 95% range of each bin under a uniform PIT. Panel titles give the median v_{t-1} over the period.

model	year	v_{t-1}			PIT u_t			
		median	IQR	mean	median	IQR	IQR width	cov. err. 50%
GEDI het.	1991	0.607	0.583–0.653	0.627	0.40	0.32–0.50	0.18	0.36
	1995	0.043	0.034–0.057	0.122	0.58	0.46–0.68	0.22	0.28
	2000	0.014	0.010–0.032	0.091	0.52	0.40–0.67	0.28	0.27
	2010	0.010	0.005–0.035	0.070	0.54	0.40–0.69	0.29	0.22
	2017	0.012	0.004–0.071	0.083	0.56	0.40–0.70	0.30	0.22
GEDI const.	1995	0.105	0.098–0.116	0.147	0.56	0.47–0.65	0.18	0.36
	2000	0.032	0.030–0.040	0.084	0.51	0.42–0.62	0.20	0.36
	2017	0.014	0.008–0.040	0.073	0.55	0.45–0.65	0.21	0.31
LandTrendr het.	1995	0.229	0.216–0.262	0.257	0.56	0.44–0.65	0.21	0.41
	2000	0.086	0.078–0.101	0.116	0.50	0.43–0.61	0.18	0.44
	2017	0.026	0.020–0.047	0.074	0.53	0.46–0.61	0.15	0.37
LandTrendr const.	1995	0.255	0.238–0.288	0.280	0.56	0.44–0.65	0.21	0.41
	2000	0.099	0.090–0.117	0.128	0.51	0.42–0.60	0.18	0.44
	2017	0.029	0.023–0.052	0.082	0.53	0.46–0.61	0.15	0.37

Table 1: Bias-corrected runs, selected years: process variance v_{t-1} ((kg C m⁻²)²) and PIT over the 538 sites. Calibrated PIT: median 0.5, IQR 0.25–0.75 (width 0.5), coverage error 0. Every year is in `fw_pit_time.csv`.

PIT (Figure 1, middle and right; Figure 2).

- **Centring: the correction works in every year.** Averaged over 1996–2017 (excluding 2003–04, see below), the corrected mean PIT is 0.54 for every model. Uncorrected it is 0.33–0.36, with median 0.27–0.32: the uncorrected forecasts are too high throughout. The remaining 0.04 above 0.5 is the low disturbance tail of the predictive (forecast-width note, Section 3).
- **Concentration: the PIT flattens only a little, and only for GEDI.** For GEDI heteroskedastic R , the PIT IQR widens from 0.22 (1995) to 0.28 (2000) and then stays at 0.29–0.30, while the median v_{t-1} falls only from 0.014 to 0.010. The central 50% interval goes from 28 to 22 points too wide. For GEDI constant R the IQR stays at 0.18–0.21. For LandTrendr it *narrows* slightly (0.21 to 0.15) even though v_{t-1} falls nine-fold: the forecast errors shrink over time faster than the forecast variance, which stays dominated by R . In Figure 2 the corrected histograms stay humped in every period.
- **The uncorrected PIT can look better, by accident.** The uncorrected 50% coverage error is near 0 or even negative for GEDI in 1995–2005. That is not calibration: the bias pushes the observations away from the centre of a predictive that is too wide, and the two errors partly cancel. The uncorrected PIT IQR is in fact narrower (0.13–0.15) than the corrected one.
- **2003–04 is a disturbance year.** Many sites lose biomass in 2003–04, which the undisturbed forecast does not expect. The PIT drops (median 0.36–0.44 corrected), its lower quartile falls to 0.04–0.20, and the coverage error dips. This is real disturbance, not better calibration, so it is excluded from the averages above.
- **A late tail toward high PIT.** From about 2010 the corrected GEDI histograms grow a tail

toward $u_t = 1$, i.e. forecasts that are too low at some sites. This matches the regrowth over-correction seen at site 300: b is learned in mature stands and is too negative for regrowing stands.

Interpretation. The process variance is the only part of the predictive variance that the filter learns, and it does shrink. But after the first few years it is a small part of the predictive variance (a median share of 1–4% of the full predictive variance from 1996 on, against 17–83% for R ; see `fw_pit_time.csv` and the forecast-width note). Once it is small, shrinking it further cannot flatten the PIT. In the R-scaling experiment (`rs_pit.png`), reducing R flattens the PIT. That points to the observation-error model, not the process variance, as the thing to change.

3 Decomposition of the predictive variance

3.1 The one-step-ahead predictive

At year t , before y_t is assimilated, the filter’s predictive for y_t is a Gaussian mixture over the τ grid points g , the two projected components $j \in \{1, 2\}$ and the disturbance classes $k \in \{0, 1, \dots, D\}$ ($k = 0$: undisturbed):

$$p(y_t \mid y_{1:t-1}) = \sum_{g,j,k} \omega_{gjk} \mathcal{N}(y_t; \mu_{gjk}, s_{gjk}^2), \quad s_{gjk}^2 = \Sigma_{gjk,33}^F + R_t. \quad (1)$$

- $\omega_{gjk} \propto \pi_{t-1}(g) w_{t-1}(g, j) \beta_{k,t}$, normalised to sum to 1. Here $\pi_{t-1}(g)$ is the τ posterior after y_{t-1} , $w_{t-1}(g, j)$ is the weight of projected component j at grid point g , and $\beta_{k,t}$ is the prior probability of class k in year t . Components whose importance weights are degenerate get $\omega = 0$.
- μ_{gjk} is the forecast mean of the wood pool: VSEM run from the previous analysis, then the disturbance draw for $k \geq 1$. With bias correction, μ_{gjk} includes the shift b_t .
- $\Sigma_{gjk,33}^F$ is the forecast variance of the wood pool. For $k = 0$ it is $\sigma_{gj0}^2 + 1/\tau_g$, where σ_{gj0}^2 is the importance-weighted ensemble variance of the VSEM forecast and $1/\tau_g$ the process variance. For $k \geq 1$ it is the ensemble variance after the disturbance draw, with no process variance added.
- R_t is the observation-error variance.

The PIT of Section 1 is $u_t = \sum_{g,j,k} \omega_{gjk} \Phi((y_t - \mu_{gjk})/s_{gjk})$, where Φ is the standard normal CDF. The predictive mean and variance are

$$m_t = \sum_{g,j,k} \omega_{gjk} \mu_{gjk}, \quad V_t = \sum_{g,j,k} \omega_{gjk} (s_{gjk}^2 + \mu_{gjk}^2) - m_t^2 \quad (\text{predVar}). \quad (2)$$

3.2 Splitting by disturbance class

Group the components by k . Class k has total weight $\beta^k = \sum_{g,j} \omega_{gjk}$ (equal to $\beta_{k,t}$ when no component is dropped), within-class weights $\omega_{gj}^k = \omega_{gjk}/\beta^k$, and within-class mean and variance

$$m^k = \sum_{g,j} \omega_{gj}^k \mu_{gjk}, \quad V^k = \sum_{g,j} \omega_{gj}^k (s_{gjk}^2 + \mu_{gjk}^2) - (m^k)^2.$$

The law of total variance gives

$$V_t = \underbrace{\beta^0 V^0}_{\text{undisturbed}} + \sum_{k \geq 1} \beta^k V^k + \sum_{k \geq 0} \beta^k (m^k - m_t)^2. \quad (3)$$

The undisturbed class is the no-disturbance predictive of the forecast-width note: $m^0 = \text{predMean0}$, $V^0 = \text{predVar0}$. Its variance splits into

$$V^0 = R_t + Q_t^0 + P_t^0, \quad Q_t^0 = \sum_{g,j} \omega_{gj}^0 / \tau_g (\text{predProcVar0}), \quad P_t^0 = \sum_{g,j} \omega_{gj}^0 \sigma_{gj0}^2 + \sum_{g,j} \omega_{gj}^0 (\mu_{gj0} - m^0)^2. \quad (4)$$

Q_t^0 is the process variance, averaged over the τ posterior. P_t^0 is the *propagated state variance*: the analysis uncertainty carried through VSEM, plus the spread of the component means. In the code $P_t^0 = V^0 - R_t - Q_t^0$.

3.3 The decomposition

Every component contains R_t , so $\sum_k \beta^k R_t = R_t$. Pulling R_t out of (3) and splitting $\beta^0 V^0$ with (4) gives an exact four-way decomposition:

$$V_t = \underbrace{R_t}_{\text{observation error}} + \underbrace{\beta^0 Q_t^0}_{\text{process error}} + \underbrace{\beta^0 P_t^0}_{\text{propagated state}} + \underbrace{D_t}_{\text{disturbance}} \quad (5)$$

$$D_t = \sum_{k \geq 1} \beta^k (V^k - R_t) + \sum_{k \geq 0} \beta^k (m^k - m_t)^2 = V_t - R_t - \beta^0 (V^0 - R_t). \quad (6)$$

- **Disturbance part D_t .** The variance from allowing a disturbance: the forecast spread of the disturbed classes, plus the spread between the class means. The undisturbed and disturbed means can be several kg C m^{-2} apart.
- **Process error $\beta^0 Q_t^0$.** This is `predProcVar`. It is the only term that contains $1/\tau$, so it is the only part of the predictive variance the τ posterior controls directly. (P_t^0 depends on τ indirectly, through earlier analyses.)
- **How it is computed.** From stored outputs: $\beta^0 = \text{predProcVar}/\text{predProcVar0}$, $R_t = \text{predR}$, and D_t from (6). The script checks that the four parts sum to V_t (to 10^{-8}).
- **Summaries.** Figure 3 shows the mean of each term over sites. Means add up, so the stack equals the mean V_t . Figure 4 shows the median over sites of each term's share of V_t . Medians do not add up, but they are not dominated by the few sites with very large Q_t^0 .
- **Data.** `predVar0` and `predProcVar0` are only stored by the 100-site subset reruns (`experiments/forecastWidth`), so this section uses those 100 sites, both runs, no product. Without the product, $\beta^0 = 0.954$, the background probability of no disturbance.

3.4 Results

- **Process error is small, and the correction makes it smaller.** With correction the mean process term falls from about 0.59 in 1991 to 0.06–0.14 from 1996 on, against 0.24–0.35 uncorrected. Its median share of V_t is 1–4%. This is the term the PIT-over-time results (Section 2) follow, and it is too small to matter by the late 1990s.

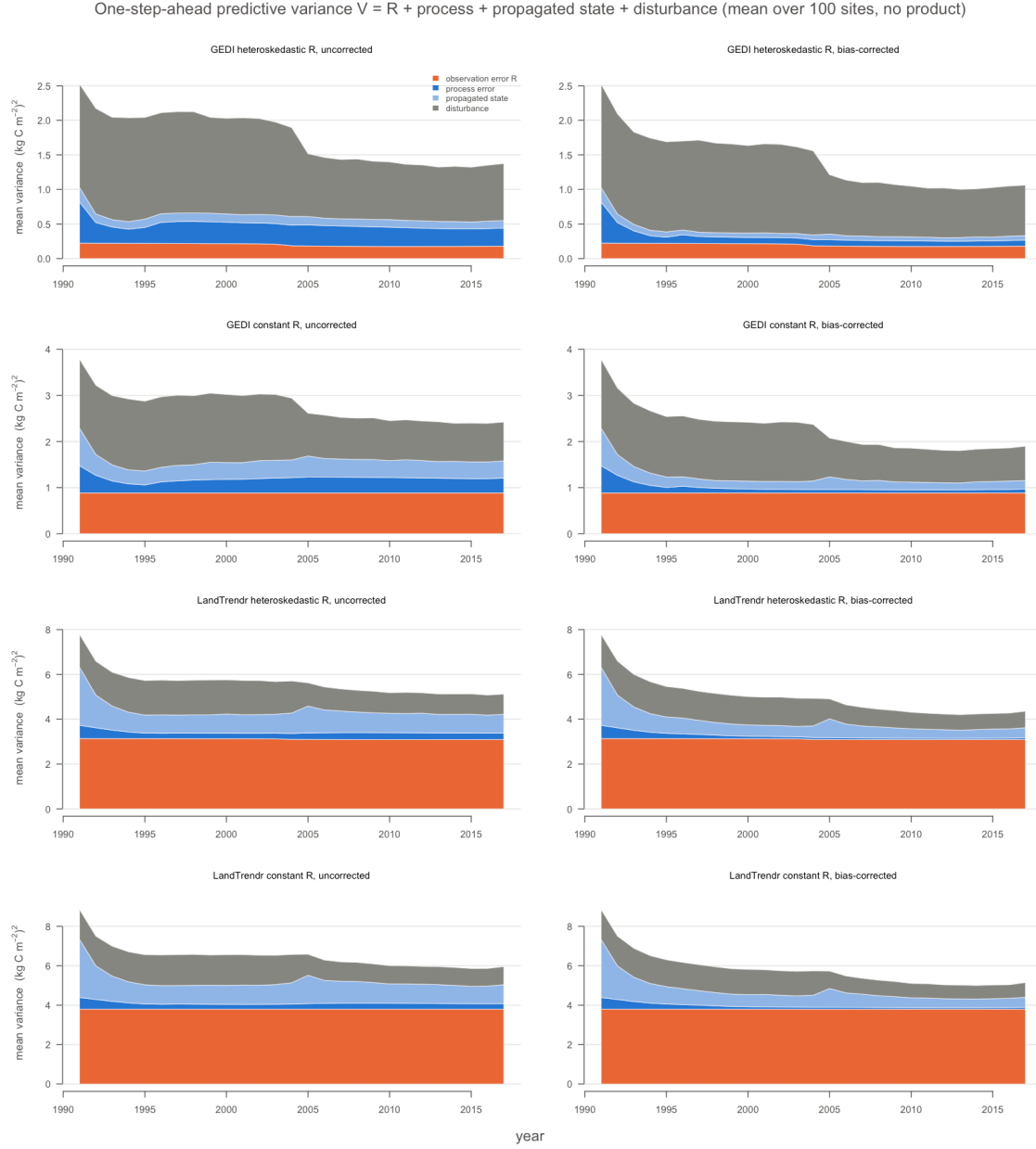


Figure 3: Decomposition (5) of the one-step-ahead predictive variance V_t , mean over the 100 sites in each year, stacked. Rows: observation-error model; columns: uncorrected, bias-corrected. Note the different y -axis ranges per row.

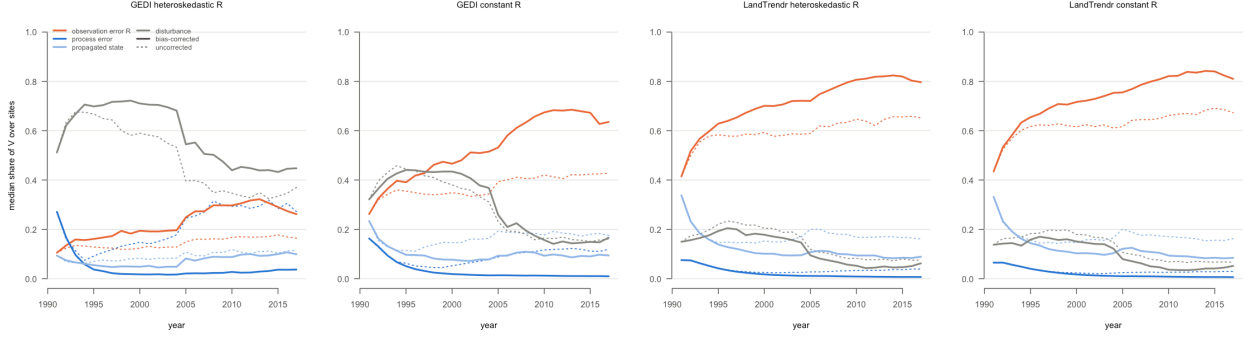


Figure 4: Median over the 100 sites of each term’s share of V_t . Solid: bias-corrected; dotted: uncorrected.

model	run	V_t	R_t	$\beta^0 Q_t^0$	$\beta^0 P_t^0$	D_t
GEDI het.	uncorr.	1.32–2.12	0.18–0.22	0.25–0.32	0.10–0.13	0.79–1.47
	corr.	1.00–1.71	0.18–0.22	0.07–0.12	0.05–0.08	0.69–1.33
GEDI const.	uncorr.	2.39–3.05	0.88	0.24–0.35	0.32–0.45	0.82–1.53
	corr.	1.80–2.56	0.88	0.07–0.14	0.15–0.27	0.70–1.32
LandTrendr het.	uncorr.	5.08–5.76	3.09–3.13	0.24–0.31	0.81–1.20	0.90–1.55
	corr.	4.20–5.38	3.09–3.13	0.06–0.22	0.36–0.84	0.69–1.32
LandTrendr const.	uncorr.	5.86–6.58	3.80	0.24–0.31	0.88–1.44	0.90–1.57
	corr.	5.00–6.17	3.80	0.08–0.23	0.44–0.95	0.68–1.33
median share, corrected						
GEDI het.			17–32%	2–4%	5–11%	43–72%
GEDI const.			42–69%	1–4%	7–11%	14–44%
LandTrendr het.			64–82%	1–3%	8–13%	4–20%
LandTrendr const.			67–84%	1–3%	8–13%	3–17%

Table 2: Range over 1996–2017 of the mean (over sites) of each term ($(\text{kg C m}^{-2})^2$), and of the median share of V_t for the corrected runs. Every year is in `fw_var_decomp.csv`.

- **R_t is fixed, so its share grows as the other terms shrink.** It does not change with correction (apart from the small y_t dependence of the heteroskedastic models). As the process and propagated-state terms shrink, its share rises: for LandTrendr from 41% (1991) to 80–84% (2010s), for GEDI constant R to 64–69%.
- **The propagated state term is large for LandTrendr.** It is 0.36–0.95 corrected, against 0.05–0.08 for GEDI heteroskedastic. With a large R the analysis barely shrinks the state uncertainty, so it is carried forward (forecast-width note, Section 3). The correction roughly halves it.
- **The disturbance part is the same for every observation-error model.** Its mean is about 1.28 before 2005 and 0.75 after, for all four. It does not depend on R : it is set by the background disturbance probability $1 - \beta^0 = 0.046$ and by how far the disturbed means sit below the undisturbed one, which scales with the biomass present. The drop after 2004 follows the 2003–04 disturbances, which left many sites with less biomass to lose. For GEDI heteroskedastic R , where every other term is small, it is the largest share of V_t (43–72%).
- **Why the disturbance part does not cause the hump** (derived in Section 3.5). D_t

is variance held by a tail with only 4.6% of the probability, several kg C m^{-2} below the undisturbed forecast. For an undisturbed year, $u_t \approx (1 - \beta^0) + \beta^0 \Phi((y_t - m^0)/\sqrt{V^0})$, so the tail only shifts the PIT up by at most 0.046 (which accounts for most of the gap between 0.5 and the 0.54 mean PIT of Section 2). How concentrated the PIT is depends on $V^0 = R_t + Q_t^0 + P_t^0$ relative to the actual errors. In that part, R_t dominates once Q_t^0 is small.

3.5 The PIT of an undisturbed year

This subsection derives the approximation used above,

$$u_t \approx (1 - \beta^0) + \beta^0 \Phi((y_t - m^0)/\sqrt{V^0}), \quad (7)$$

starting from the exact PIT of the mixture predictive (1),

$$u_t = \sum_{g,j,k} \omega_{gjk} \Phi((y_t - \mu_{gjk})/s_{gjk}), \quad (8)$$

for a year in which no disturbance happened.

Step 1: group by disturbance class. With $\omega_{gjk} = \beta^k \omega_{gj}^k$ (Section 3.2), (8) becomes a weighted sum of the class CDFs evaluated at y_t :

$$u_t = \sum_{k \geq 0} \beta^k F^k(y_t), \quad F^k(y) = \sum_{g,j} \omega_{gj}^k \Phi((y - \mu_{gjk})/s_{gjk}). \quad (9)$$

F^k is the CDF of the predictive given $z_t = k$. So far this is exact.

Step 2: the disturbed classes lie below the observation. In an undisturbed year, y_t is near the undisturbed forecast m^0 . The disturbed component means μ_{gjk} , $k \geq 1$, are several kg C m^{-2} below it, because a disturbance removes a large fraction of the wood pool. So $(y_t - \mu_{gjk})/s_{gjk}$ is large and positive, and $F^k(y_t) \approx 1$ for every $k \geq 1$:

$$\sum_{k \geq 1} \beta^k F^k(y_t) \approx \sum_{k \geq 1} \beta^k = 1 - \beta^0. \quad (10)$$

The size of the error is bounded whether or not the disturbed classes really are far below. Since $0 \leq F^k \leq 1$,

$$\beta^0 F^0(y_t) \leq u_t \leq (1 - \beta^0) + \beta^0 F^0(y_t), \quad (11)$$

so replacing the disturbed part by $1 - \beta^0$ overstates u_t by at most $1 - \beta^0$ (0.046 without the product). It is exact when all the disturbed mass lies below y_t .

Step 3: approximate the undisturbed class by one Gaussian. F^0 is the CDF of a Gaussian mixture over (g, j) , with mean m^0 and variance V^0 (Section 3.2). Replacing it with the CDF of a single Gaussian with the same two moments,

$$F^0(y_t) \approx \Phi((y_t - m^0)/\sqrt{V^0}), \quad (12)$$

is accurate near the centre of the distribution, which is where an undisturbed observation usually lands. It is less accurate in the tails: a mixture over τ has heavier tails than one Gaussian with the same variance. It is exact when one component carries all the weight.

Result. Substituting (10) and (12) into (9) gives (7).

Check. On the bias-corrected 100-site subset (no product), over the 10 323 undisturbed site-years ($t \geq 3$, $P(z_t \neq 0 \mid y_{1:t}) < 0.5$, all four observation-error models), (7) computed from `predMean0`, `predVar0` and β^0 agrees with the stored exact PIT (8) (`hist$pit`). The mean absolute difference is 0.0085, 95% of differences are below 0.022, the largest is 0.077, and the correlation is 0.999. The approximation is on average 0.008 too high, as (11) predicts when a little disturbed mass lies above y_t . The IQRs are 0.191 (approximation) and 0.198 (exact).

What it implies.

- **The disturbance tail only shifts the PIT.** Write $z_t = (y_t - m^0)/\sqrt{V^0}$. The tail adds a constant $1 - \beta^0$ and compresses $\Phi(z_t)$ into $[1 - \beta^0, 1]$. It cannot change the *shape* of the PIT histogram, only move it right by up to 0.046 and squeeze it by the factor $\beta^0 = 0.954$. If the tail were removed, the mean PIT would fall by about $(1 - \beta^0)(1 - \mathbb{E}[\Phi(z_t)]) \approx 0.02$ (0.54 to 0.52), and the IQR would grow by a factor $1/\beta^0 \approx 1.05$.
- **The hump is set by the undisturbed forecast.** The shape comes from $\Phi(z_t)$, which is uniform only if z_t is standard normal, i.e. if $V^0 = R_t + Q_t^0 + P_t^0$ matches the actual squared errors. If V^0 is too large, z_t is concentrated near 0 and $\Phi(z_t)$ near 0.5. With the forecast-width note’s median $|z_t| = 0.39$ (against 0.674 for a standard normal), $\Phi(z_t)$ has quartiles of about 0.35 and 0.65, an IQR of about 0.30, close to the observed PIT IQR for GEDI heteroskedastic R .
- **Variance and PIT weight the tail differently.** V_t counts squared distance and the PIT counts probability. With undisturbed spread $\sqrt{V^0} \approx 0.6$ and a gap of about 5 kg C m^{-2} to the disturbed means, the tail adds roughly $\beta^0(1 - \beta^0) \cdot 5^2 \approx 1.1$ to V_t , close to the D_t in Table 2. That is about three times V^0 , but in the PIT it amounts to only the constant 0.046.
- **Disturbance years are the exception.** If a disturbance did happen, y_t falls inside or below the disturbed classes. Then $F^k(y_t) < 1$, Step 2 fails, and u_t can be near 0 (the spike near 0 in 2001–05 in Figure 2).

Reproducing

Rscript `claude-working-docs/forecast_width_analysis.R`, from the repository root, with `EXP_DIR` set if the experiment folders are elsewhere. Section 6 of the script writes `fw_pit_time.png`, `fw_pit_periods.png` and `fw_pit_time.csv`. Section 7 writes `fw_var_decomp.png`, `fw_var_share.png` and `fw_var_decomp.csv`, and prints the check of Section 3.5 (“7b”); it uses the 100-site subset reruns in `experiments/forecastWidth`, which section 2 creates if they are missing.